

Presentation title

placeholder - final title page to be added later

Compression pipeline + no-reference IQMs

JPEG û JPEG2000 û JPEG XR û JPEG XL û JPEG AI / BRISQUE û NIQE û NIMA û CLIP-IQA

Compression experiment: workflow and objective

Task performed

- ▶ Selected representative **600 × 600** test images.
- ▶ Compressed each image with **JPEG, JPEG2000, JPEG XR, JPEG XL, JPEG AI**.
- ▶ Used fixed target filesizes: **175, 150, 125, 100, 75, 50, 25 kB**.
- ▶ Assessed all outputs with **BRISQUE, NIQE, NIMA, CLIP-IQA**.

Why this was done

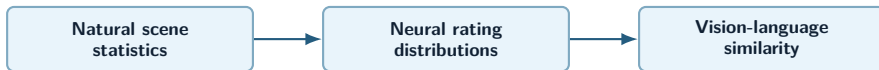
- ▶ Identify suitable codecs for different byte budgets.
- ▶ Study quality changes under strong compression.
- ▶ Check whether no-reference IQMs capture compression artefacts.
- ▶ Support codec comparison without reference images.



At small filesizes, visible compression artefacts are expected; the IQMs should reflect these degradations.

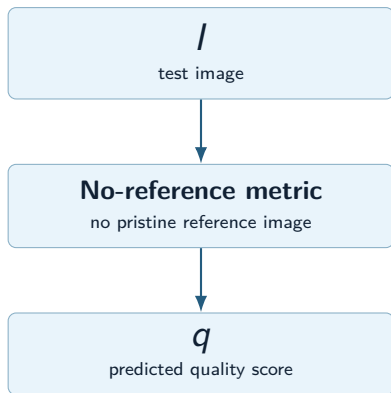
BRISQUE, NIQE, NIMA, CLIP-IQA

Natural statistics, rating distributions and vision-language similarity



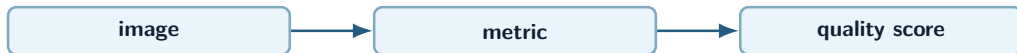
Primary sources: Mittal et al. 2012; Mittal et al. 2013; Talebi and Milanfar 2018; Wang et al. 2023

Common framing: no-reference IQA

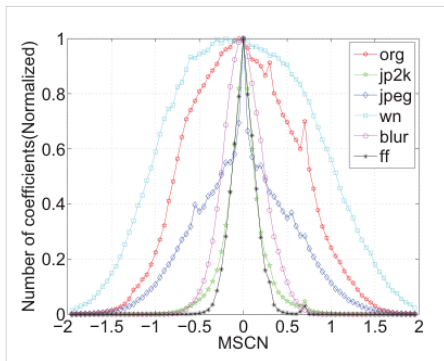


Scope

- ▶ Image quality assessment predicts perceived image quality.
- ▶ No-reference IQA uses only the test image.
- ▶ BRISQUE and NIQE use natural scene statistics.
- ▶ NIMA uses human rating distributions.
- ▶ CLIP-IQA uses image-text similarity.



BRISQUE: core idea



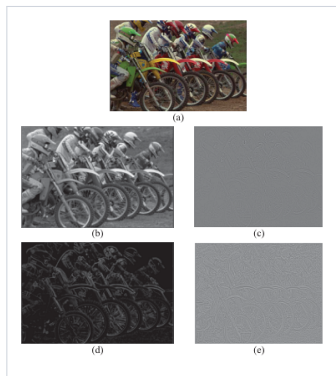
Source: Mittal, Moorthy, and Bovik (2012), Fig. 4.

Method facts

- ▶ Works directly in the spatial domain.
- ▶ Uses locally normalized luminance coefficients.
- ▶ Models MSCN coefficients with a GGD.
- ▶ Models neighboring MSCN products in four directions.
- ▶ Maps features to quality using SVR.



BRISQUE: formula and pipeline



MSCN coefficient

$$\hat{l}(i,j) = \frac{l(i,j) - \mu(i,j)}{\sigma(i,j) + C}$$

- ▶ $l(i,j)$: luminance value.
- ▶ $\mu(i,j)$: local mean.
- ▶ $\sigma(i,j)$: local contrast.
- ▶ $\hat{l}(i,j)$: MSCN coefficient.
- ▶ C : stabilizing constant.

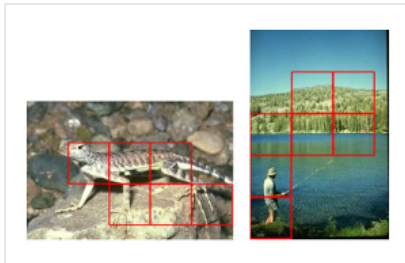
Neighbor products

$$H = \hat{l}(i,j)\hat{l}(i,j+1), \quad V = \hat{l}(i,j)\hat{l}(i+1,j)$$

Source: Mittal, Moorthy, and Bovik (2012), Fig. 3.



NIQE: core idea



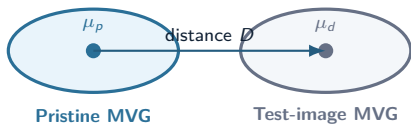
Method facts

- ▶ NIQE is called “completely blind.”
- ▶ It uses no human-rated distorted images.
- ▶ It uses no exposure to distorted images.
- ▶ It learns NSS features from natural images.
- ▶ Quality is distance to the natural MVG model.

Source: Mittal, Soundararajan, and Bovik (2013), Fig. 1.



NIQE: formula and pipeline

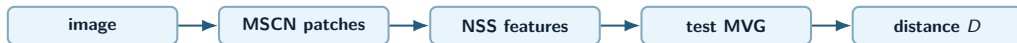


compare the test-image MVG with
the pristine natural-image MVG

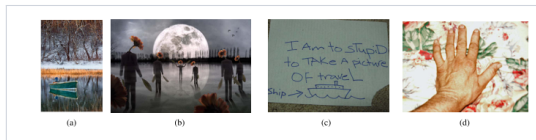
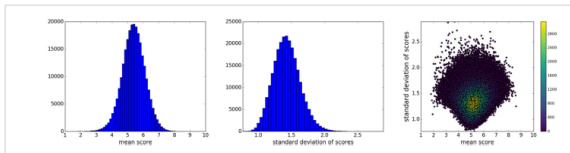
NIQE distance

$$D = \sqrt{(\mu_p - \mu_d)^T \left(\frac{\Sigma_p + \Sigma_d}{2} \right)^{-1} (\mu_p - \mu_d)}$$

- ▶ μ_p, Σ_p : pristine natural-image MVG.
- ▶ μ_d, Σ_d : test-image MVG.
- ▶ D : NIQE distance score.
- ▶ Lower distance means closer to pristine statistics.



NIMA: core idea



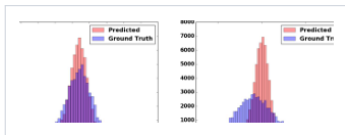
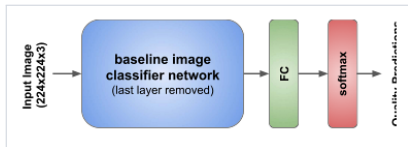
Method facts

- ▶ NIMA uses a convolutional neural network.
- ▶ It predicts a distribution of human scores.
- ▶ It does not predict only the mean score.
- ▶ It uses no golden reference image.
- ▶ It supports no-reference perceptual assessment.

Source: Talebi and Milanfar (2018), Figs. 1–2.



NIMA: formula and pipeline



Source: Talebi and Milanfar (2018), Figs. 8–9.

Distribution output

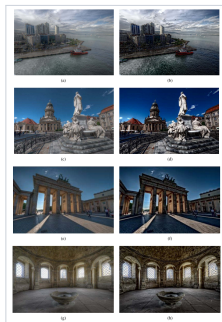
$$\hat{p}_i = \frac{e^{z_i}}{\sum_{j=1}^N e^{z_j}}, \quad \hat{\mu} = \sum_{i=1}^N s_i \hat{p}_i$$

$$\hat{\sigma}^2 = \sum_{i=1}^N (s_i - \hat{\mu})^2 \hat{p}_i$$

- ▶ z_i : network logit.
- ▶ $\hat{p}_i$: predicted score probability.
- ▶ s_i : rating value.
- ▶ $\hat{\mu}$: predicted mean score.
- ▶ $\hat{\sigma}^2$: predicted score variance.



NIMA: training loss

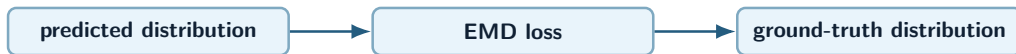


EMD facts

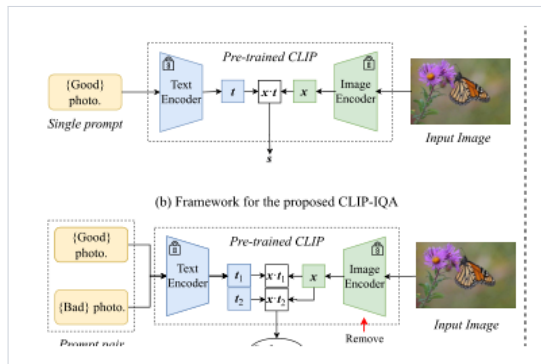
- ▶ Ratings are ordered classes.
- ▶ The model predicts score probabilities.
- ▶ Training compares predicted and true distributions.
- ▶ NIMA uses Earth Mover's Distance.
- ▶ EMD measures distribution closeness.

$$EMD(p, \hat{p}) = \left(\frac{1}{N} \sum_{k=1}^N |CDF_p(k) - CDF_{\hat{p}}(k)|^r \right)^{1/r}$$

Source: Talebi and Milanfar (2018), Fig. 14.



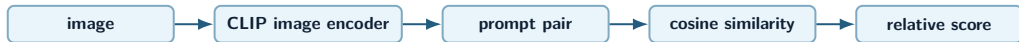
CLIP-IQA: core idea



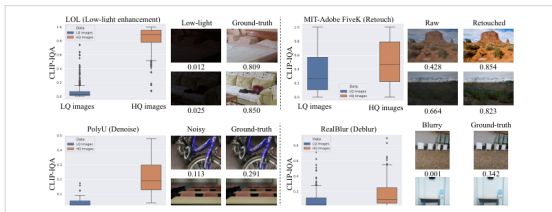
Method facts

- ▶ CLIP embeds images and text.
- ▶ Single prompts are affected by linguistic ambiguity.
- ▶ CLIP-IQA uses antonym prompt pairs.
- ▶ Example: “Good photo.” / “Bad photo.”
- ▶ The final score is relative similarity.

Source: Wang, Chan, and Loy (2023), Fig. 2.



CLIP-IQA: formula and pipeline

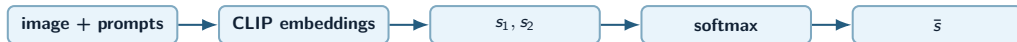


Source: Wang, Chan, and Loy (2023), Fig. 5.

Prompt-pair score

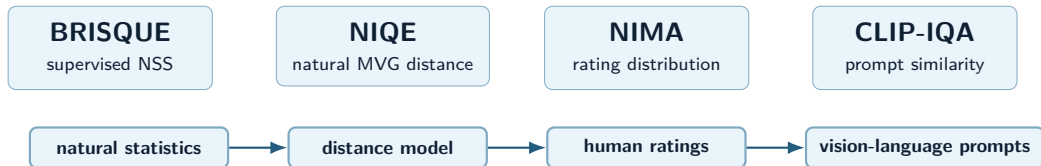
$$s_i = \frac{x \cdot t_i}{\|x\| \|t_i\|}, \quad \bar{s} = \frac{e^{s_1}}{e^{s_1} + e^{s_2}}$$

- x : image feature vector.
- t_i : text prompt feature vector.
- s_i : cosine similarity.
- $\bar{s}$: final CLIP-IQA score.
- T_1, T_2 : antonym prompt pair.



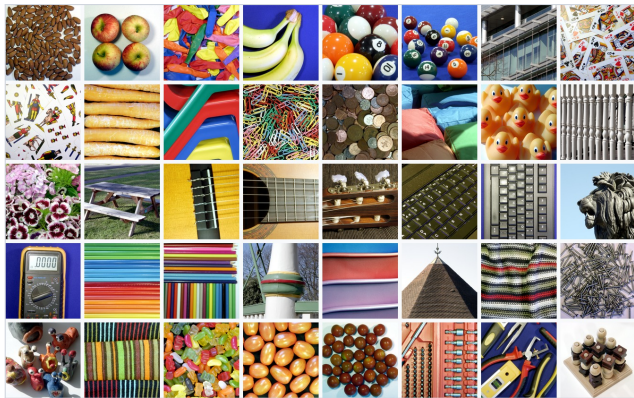
Score ranges and direction

Metric	Output used in the presentation	Direction	Range / note
BRISQUE	SVR quality score, trained against human opinion scores / DMOS	lower is better	commonly [0,100]; paper correlations are with DMOS
NIQE	distance between pristine MVG and test-image MVG	lower is better	no fixed upper bound; 0 would mean identical MVG means
NIMA	expected value of predicted rating distribution	higher is better	dataset-dependent: AVA/LIVE [1,10], TID2013 [0,9]
CLIP-IQA	softmax relative similarity to positive prompt T_1	higher is better for positive T_1	[0, 1]



Sources: four primary papers. BRISQUE/NIQE practical score interpretation cross-checked with MathWorks documentation.

Test dataset: TESTIMAGES 600×600 samples



Dataset role

- ▶ Natural scenes with varied textures, edges and colors.
- ▶ Used subset: **600×600 RGB** samples.
- ▶ Same images are available at multiple resolutions.
- ▶ Enables the same compression/IQM experiment at larger file sizes.

Source: testimages.org. Citation: Asuni, N., & Giachetti, A. (2015). *TESTIMAGES: A Large Data Archive for Display and Algorithm Testing*. *Journal of Graphics Tools*, 17(4), 113–125. DOI: 10.1080/2165347X.2015.1024298.

What the src/ folder contains

main.py

Encodes images to target byte sizes.

eval.py

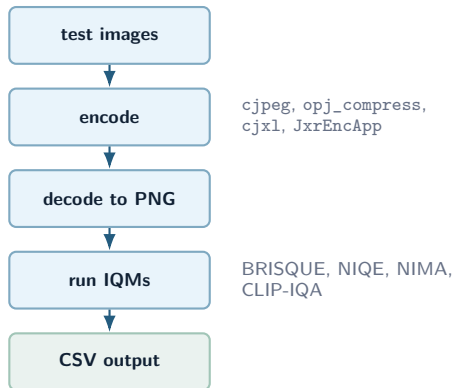
Runs IQMs on decoded PNG folders.

read_to_df.py

Extracts metadata + metrics into a DataFrame.

data.csv / metrics.csv

Tabular outputs for analysis.



Libraries and external tools

Python packages

typer

tqdm

pathlib

pandas

pyiqa

re

subprocess

Callable

Role split

subprocess calls codecs; pyiqa creates no-reference metrics; pandas stores rows.

Codec tools / parameter modes

JPEG

`cjpeg -quality q`

JPEG2000

`opj_compress -r r`

JPEG XL

`cjxl -q q`

JPEG XR

`JxrEncApp -q q`

JPEG AI

`estimated q`

How target sizes are defined

```
FORMATS = ['jpeg', 'j2k', 'jxl', 'jxr']  
SIZES = [25000, 50000, 75000,  
         100000, 125000, 150000, 175000]
```



Naming convention

Compressed files are written as:

`format_target_image`
`.format`

Interpretation

Each image is encoded once for every codec and every fixed byte target.

How target sizes are acquired

A) Iterative quality search

Encode, measure bytes, then narrow the quality interval.

```
lo, hi = min_q, max_q
while lo + 1 < hi:
    q = (lo + hi) // 2
    bytes = encode(q)
    update_bounds(bytes, target)

encode(best_q)
```

B) Pre-estimated parameter

For JPEG AI, the actual quality parameter was sometimes estimated before encoding.

```
q = estimate_quality(image, target)
bytes = encode_jpeg_ai(q)

if not close(bytes, target):
    refine q or rerun encode
```

Shared goal: select a codec-specific parameter so the encoded image lands near the fixed byte target.

No-reference IQMs in the code

```
IQMS = {  
    'brisque': pyiqa.create_metric('brisque'),  
    'niqe':    pyiqa.create_metric('niqe'),  
    'nima':    pyiqa.create_metric('nima'),  
    'clip':    pyiqa.create_metric('clipiqa')  
}  
  
score = metric(str(img_path)).item()
```

BRISQUE

natural-scene stats

NIQE

naturalness model

NIMA

aesthetic score

CLIP-IQA

semantic quality

Applied to decoded PNG images - no reference image required.

Evaluation loop and CSV rows

```
for fmt in FORMATS:
    for size in SIZES:
        for img in decoded_folder.glob('*.png'):
            row = {
                'image_id': img.stem,
                'format': fmt,
                'target_size': size
            }
            for name, metric in IQMS.items():
                row[name] = metric(str(img)).item()
            data_rows.append(row)
```

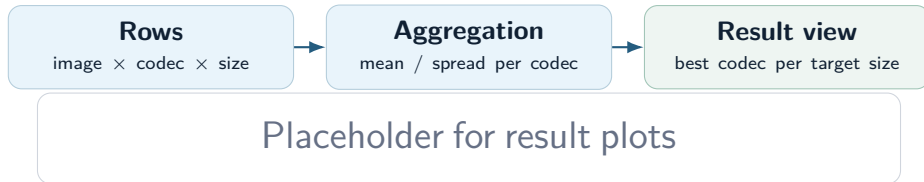
One row = one evaluated image

image_id	source image
format	codec
target_size	byte target
brisque...clip	IQM scores

Output

metrics.csv: compact IQM table.
data.csv: metadata + IQMs.

Results: codec comparison



Results: IQM behavior

Placeholder

BRISQUE / NIQE trends

Placeholder

NIMA / CLIP-IQA trends

Results: takeaway placeholder

